Africa-Asia Biosciences Challenge Fund (AABCF) Eastern African Workshop: Advancing Climate-Smart Livestock Practices Through Science and Partnership

Project Name: **Africa-Asia Biosciences Challenge Fund (AABCF)**

Date and venue: *ILRI campus, Nairobi, Kenya, 19-23 January 2026*

Flagship Report:

Africa-Asia Biosciences Challenge Fund (AABCF) Stakeholder Consultation Workshop:

Advancing Climate-Smart Livestock through Science, Innovation, and Partnership

1. Executive Summary

Livestock systems across Africa and Southeast Asia are at the center of food security, livelihoods, and climate resilience, yet face escalating pressures from climate variability, heat stress, disease, land degradation, and rising demand for animal-source foods. At the same time, livestock, particularly ruminants, are increasingly scrutinized for their contribution to greenhouse gas (GHG) emissions, especially methane.

The Africa-Asia Biosciences Challenge Fund (AABCF) Eastern Africa Workshop, convened in Nairobi from 19-23 January 2026, brought together 64 participants from Africa, Southeast Asia, Australia, CGIAR centers, universities, and national research institutions. Hosted by the International Livestock Research Institute (ILRI) in partnership with the University of Queensland and the University of Melbourne, with support from the Australian Centre for International Agricultural Research (ACIAR) and the Ballmer Group, the workshop combined technical training, policy dialogue, field-based learning, and research showcases.

The workshop demonstrated that climate-smart livestock (CSL) systems are achievable through integrated, productivity-led approaches that simultaneously enhance resilience and reduce emissions intensity. Improving animal health, feed quality, genetics, and data systems emerged as the most cost-effective and scalable pathways to climate mitigation and adaptation in low- and middle-income contexts. The event strengthened Africa-Asia research networks, enhanced technical capacity, and generated clear policy- and donor-relevant investment priorities. A similar workshop will be organized in Vietnam to tailor to the SEA Livestock context.

2. Background and Rationale

Livestock systems in Africa and Southeast Asia support hundreds of millions of households, particularly in arid and semi-arid lands (ASALs). These systems are increasingly exposed to climate shocks, including prolonged droughts, rising temperatures, floods, and dynamic shifting diseases. Simultaneously, rapid population growth and urbanization are driving increased demand for animal-source foods, placing further pressure on fragile production systems.

While livestock contribute significantly to national economies and nutrition, they are also a major source of methane emissions. This dual role creates an urgent need for solutions that improve productivity and livelihoods while reducing environmental footprints. The AABCF workshop was designed to respond to this challenge by strengthening scientific capacity, fostering cross-regional collaboration, and accelerating the translation of research into policy and practice.

3. Workshop Objectives and Design

The workshop pursued four interlinked objectives:

- Establish a shared conceptual foundation for climate-smart livestock systems, emphasizing productivity, resilience, and mitigation.

- Strengthen technical capacity in animal health, nutrition, genetics, grazing management, methane mitigation, and data systems.

- Demonstrate applied, real-world CSL systems through field-based learning in semi-arid rangelands.

- Catalyze Africa-Asia and South-South research collaboration and improve research-to-policy and research-to-practice translation.

The five-day program combined expert-led lectures, roundtable discussions, hands-on demonstrations, a field visit, and presentations by AABCF Fellows.

4. Core Scientific and Technical Insights

4.1 Climate-Smart Livestock as an Integrated Systems Approach

A central message of the workshop was that CSL is not defined by single technologies, but by integrated solutions addressing feed, animal health, genetics, management, environment, and markets simultaneously. Productivity gains emerged as the most immediate and practical pathway to reducing emissions intensity, particularly in extensive and smallholder systems.

Animal health was repeatedly highlighted as a high-impact climate solution. Disease reduces productivity, increases herd sizes required to meet demand, and raises emissions per unit of output. Improving animal health therefore delivers a “triple win”: increased productivity, enhanced resilience, and lower emissions intensity.

4.2 Climate Stress, Heat, and Animal Performance

Heat stress was identified as a unifying risk across regions. Rising temperatures reduces feed intake, compromise gut integrity, increases oxidative stress, and leads to substantial losses in milk, meat, and reproductive performance. Effective responses require combining short-term measures (shade, cooling, nutritional supplementation) with long-term genetic selection for heat tolerance.

4.3 Nutrition, Grazing, and Methane Mitigation

Methane emissions represent both an environmental cost and a direct loss of animal energy. In low-input systems, the priority is reducing emissions intensity rather than absolute emissions. The workshop emphasized immediately deployable mitigation options, including:

- Improving forage quality and digestibility
- Integrating legumes and alternative feeds
- Strategic supplementation with oils and nitrates
- Optimizing grazing pressure and stocking rates

High-tech options such as inhibitors and vaccines were presented as promising future complements but not substitutes for productivity-led strategies.

4.4 Genetics and Data as Enablers of Resilience

Indigenous and locally adapted breeds consistently outperformed exotic breeds under heat, disease, and low-quality feed conditions. Genomic tools offer opportunities to accelerate improvement while maintaining adaptation, particularly when combined with within-breed selection and participatory approaches.

Robust data systems were identified as a critical constraint. Adoption of FAIR data principles, improved livestock censuses, and integration of digital and spatial tools are essential for performance monitoring, emissions accounting, early warning systems, and evidence-based policy.

5. Five-Day Workshop Synthesis

Day 1- Foundations of Climate-Smart Livestock

The opening day established the CSL framework and highlighted a policy shift from drought relief toward livestock asset protection and market-oriented resilience. Technical sessions demonstrated that animal health and productivity gains are central to climate mitigation and that data-driven decision-making underpins effective livestock planning.

Day 2 - Nutrition, Genetics, and Management

Day 2 focused on practical biological and management solutions to reduce emissions while improving productivity. Indigenous breeds, improved forages, and grazing management were emphasized as scalable options. Country roundtables identified priority challenges, research gaps, and adoption strategies.

Day 3 - Field-Based Learning at Kapiti Research Station

Participants visited Kapiti Research Station and Wildlife Conservancy, a 13,000-hectare semi-arid landscape integrating livestock research, production, and wildlife conservation. The visit demonstrated climate-smart feeds, animal health management, genetic improvement, methane measurement, and landscape-scale resilience. The field visit also illustrated how CSL solutions operate under real-world conditions. Integrated approaches to feeds, animal health, genetics, methane measurement, and rangeland management were demonstrated within a wildlife–livestock interface. The visit highlighted the importance of community incentives, policy support, and landscape-scale planning for sustainable resilience.

Photo 1: Field-Based Training and Participant Induction at Kapiti Research Station

Day 4 - Innovation, Data, and Measurement

Sessions introduced GIS, earth observation, and predictive modeling for disease and rangeland management. A visit to the Mazingira Centre provided hands-on exposure to GHG measurement technologies and reinforced the importance of credible data.

Day 5 - Fellows' Research and Strategic Wrap-Up

AABCF Fellows showcased practical, science-based innovations demonstrating that climate-smart livestock solutions can be locally developed, cost-effective, and scalable across Africa and Southeast Asia.

Key findings highlighted productivity-led methane mitigation as the most immediate climate lever. A joint Kenya-Ethiopia study combining a locally developed probiotic (KalPro) with indigenous seaweeds achieved 51-56% methane reduction in vitro, while maintaining milk production under drought conditions and increasing propionate production by over 100%, indicating improved energy efficiency rather than trade-offs between productivity and mitigation.

Several presentations advanced circular economy approaches, including Black Soldier Fly larvae reared on sanitized human waste for poultry feed, and composting slaughterhouse waste to replace synthetic fertilizers, both addressing emissions, waste management, and input costs. One Health considerations, particularly antimicrobial resistance and biosafety, were explicitly integrated into these innovations.

Digital and genomic tools emerged as key enablers of resilience. High-resolution satellite data and machine learning predicted rangeland biomass with over 90% accuracy, supporting early warning systems and conflict reduction over grazing resources. Genomic analyses of indigenous poultry across Africa identified candidate genes for heat and disease tolerance, enabling productivity gains without loss of adaptation.

Across regions, fellows emphasized locally adapted breeds, alternative feeds, and data-driven management as foundational climate-smart investments. Collectively, the findings demonstrate that Africa and Asia can design, test, and scale their own climate-smart livestock solutions, delivering simultaneous gains in productivity, resilience, and emissions reduction.

Discussions emphasized the need for longer fellowships, stronger policy engagement, and sustained collaboration.

6. Africa-Asia Collaboration and Innovation

Cross-regional exchange among African and Southeast Asian researchers revealed strong convergence around shared constraints, including feed scarcity, disease pressure, climate risk, and weak adoption pathways. Fellow-led research demonstrated the effectiveness of locally adapted solutions, spanning circular economy innovations, genomics, and digital decision-support tools, reinforcing the value of South–South and triangular collaboration. The workshop also strengthened communication and presentation skills, particularly in a multilingual setting where language barriers remain a challenge. Key recommendations include investing in sustained Africa-Asia research platforms, embedding communication and science-to-policy skills in capacity-building programs, and prioritizing co-designed, locally grounded innovations to accelerate adoption and impact at scale. A similar workshop will be organized in Vietnam to tailor to the SEA Livestock context.

7. Roundtable Synthesis

7.1. Major Findings and Recommendations

Country roundtable discussions from Tanzania, Ethiopia, Kenya, and Southeast Asia revealed strong convergence around the core constraints facing climate-smart animal agriculture. Short-term challenges are dominated by feed shortages (quantity and quality), high disease burden and weak biosecurity, limited water availability, and poor livestock data systems. Long-term challenges cut across countries and include weak characterization and utilization of indigenous breeds, low adoption of climate-smart technologies (including digital tools), persistent environmental degradation, and inadequate institutional and human capacity to support climate-smart livestock transitions.

Participants emphasized a clear research priority agenda, centered on developing low-methane, high-productivity feeds and breeds, improving disease surveillance and diagnostic capacity, strengthening understanding of national livestock populations, and generating locally relevant emission factors. Across all countries, the need for context-specific evidence, rather than imported assumptions, was strongly highlighted.

On research translation and adoption, the roundtables consistently stressed that technology alone is insufficient. Effective uptake requires bundled solutions (feed, health, breeds, and management), strengthened extension systems, farmer field schools, climate-smart villages, and sustained engagement between researchers, policymakers, and producers. Digital tools, media, early warning systems, and insurance were identified as critical enablers of scale and resilience.

Overall recommendation: shift from fragmented interventions to integrated, productivity-led, and farmer-centered climate-smart livestock strategies, supported by strong data systems, policy alignment, and long-term capacity investment.

7.2 Policy Implications and Research Recommendations

7.2.1 Policy Implications for Africa and Asia

Policy implication for both continents includes:

- Mainstream climate-smart livestock (CSL) as a core pillar of national climate, food security, and economic transformation strategies, particularly in arid and semi-arid lands (ASALs).
- Shift public investment frameworks from short-term drought relief toward livestock asset protection, productivity enhancement, and market-oriented resilience mechanisms.
- Integrate animal health explicitly into Nationally Determined Contributions (NDCs) and livestock climate strategies as a cost-effective mitigation and adaptation lever.
- Promote data-driven livestock planning through strengthened livestock censuses, earth observation tools, and digital decision-support systems tailored to mobile pastoral systems.
- Prioritize indigenous and locally adapted breeds in breeding policies while supporting structured cross-breeding programs that balance productivity and resilience.
- Incentivize low-emission feeding strategies (improved forages, legumes, circular feeds) through extension services, subsidies, and climate finance mechanisms.
- Adopt a One Health policy approach that links livestock, wildlife, and human health to manage emerging diseases under climate change.
- Strengthen regional and South–South research partnerships to accelerate technology transfer, harmonize standards, and reduce duplication of effort across Africa and Asia.

7.2.2 Recommendations for Further Research

Recommendations for further research include:

- Develop scalable metrics linking animal health improvements to productivity gains and greenhouse gas emission reductions in smallholder and pastoral systems.
- Conduct long-term field validation of methane mitigation strategies (legumes, probiotics, oils, nitrates) under diverse African and Asian production environments.
- Advance genomic and phenotypic research on heat tolerance, disease resistance, and feed efficiency in indigenous livestock breeds.
- Investigate climate-disease-nutrition interactions, including vector-borne disease dynamics and antimicrobial resistance pathways under changing climates.
- Expand research on small ruminants and poultry as climate-resilient livestock options, with emphasis on gender, youth, and low-resource households.
- Assess the socio-economic and policy feasibility of circular economy innovations (insect protein, waste composting, fermented bedding) at scale.
- Strengthen early warning systems by integrating earth observation, climate forecasts, and mobile livestock movement data for drought and conflict prevention.
- Evaluate institutional and financing models that accelerate adoption of climate-smart livestock innovations by smallholders and pastoralists.

7.2.3 Executive workshop Evaluation Summary

The Africa-Asia Biosciences Challenge Fund (AABCF) Eastern Africa Workshop was effectively delivered and highly valued by participants. Workshop logistics and coordination were strong, with smooth operations, good time management, and high-quality technical presentations. Participant selection was a key strength, enabling cross-country, cross-disciplinary learning and productive exchange among researchers from Africa, Southeast Asia, Australia, CGIAR centers, and national institutions. The workshop successfully fostered networking, collaboration, and peer learning, particularly among AABCF Fellows, and addressed topics that were highly relevant to climate-smart livestock (CSL) priorities.

The multidisciplinary structure, integrating animal health, nutrition, genetics, grazing management, methane mitigation, and data systems, was especially effective. Roundtable discussions and field-based learning at ILRI and Kapiti Research Station reinforced systems thinking and helped participants identify future research priorities and collaboration opportunities. Overall, the workshop met its core objectives of capacity strengthening, knowledge exchange, and partnership building.

Areas for improvement relate primarily to depth, sequencing, and continuity. The workshop duration was insufficient given the breadth of content, and Fellows' presentations would benefit from earlier scheduling to allow more feedback and mentorship. Greater engagement of policymakers, industry actors, and donors is needed to strengthen research translation and scaling pathways. Additional attention is required for scientific writing, data analysis skills, practical hands-on sessions, and improved communication around logistics and access to materials. Broader livestock system coverage, including poultry and small livestock, would further enhance relevance.

Key lessons learned include the value of sustained post-workshop engagement (e.g. webinars and follow-up meetings), longer fellowship support, research and effort oriented based on country priorities and clearer pathways linking research outputs to policy and development impact. Overall, the workshop demonstrated strong value for investment and provides a solid foundation for scaling climate-smart livestock research through extended capacity building to include soft skills, enterprise development, stronger partnerships, and targeted follow-on support.

8. Value Proposition for ACIAR

The workshop demonstrated strong alignment with ACIAR priorities:

- Productivity-led mitigation delivers rapid, scalable impact.
- Animal health is a high-return, low-cost climate investment.
- Practical methane mitigation options are already available.
- Digital and spatial tools offer scalable public goods.
- South-South collaboration increases return on research investment.

Australian scientific expertise in genetics, nutrition, methane mitigation, and systems analysis proved highly relevant to African and Asian contexts, reinforcing the value of continued partnership.

This flagship report is explicitly framed to align with ACIAR's Investment Framework and Theory of Change, emphasizing:

- Productivity-led growth as the primary pathway to poverty reduction and food security.
- Climate resilience and adaptation in smallholder and extensive livestock systems.
- High-return, context-appropriate agricultural R&D with clear adoption pathways.
- Capacity building and enduring partnerships across Africa, Asia, and Australia.
- Impact-oriented research translation, from evidence generation to policy uptake and producer adoption.

Photo 2: Representatives of Australian Partner Institutions Supporting the Workshop

9. Conclusions and Way Forward

The AABCF Eastern Africa Workshop demonstrated that climate-smart livestock systems are achievable through integrated, context-specific, and science-based approaches. Productivity, resilience, and mitigation are mutually reinforcing objectives when addressed holistically.

Sustained engagement beyond short-term events, through longer fellowships, continuous learning platforms, and stronger policy and industry linkages, will be essential to translate capacity building into large-scale systems transformation. Investing in climate-smart livestock represents both an environmental imperative and a strategic opportunity to support livelihoods, food security, and inclusive economic development across Africa and Southeast Asia.

10. Acknowledgment

The organizers of the AABCF Stakeholder Consultation Workshop, held in Nairobi, Kenya, from 19-23 January 2026, extend their sincere appreciation to all participants and partner institutions who contributed their time, expertise, and valuable insights to the consultation process. The workshop brought together representatives from universities, national research organizations, government ministries, international research institutes, development partners, and AABCF fellows from Africa and Asia. Their active engagement and thoughtful contributions were instrumental in shaping the vision, priorities, and collaborative pathways for the AABCF initiative.

We gratefully acknowledge the participation and support of representatives from the following institutions:

- Haramaya University
- Jomo Kenyatta University of Agriculture and Technology
- Tanzania Livestock Research Institute
- University of Dodoma
- Bio and Emerging Technology Institute
- Kenya Agricultural and Livestock Research Organization
- University of Nairobi
- Kenya Meat Commission
- Kenyatta University
- University of Melbourne
- University of Queensland
- Ministry of Agriculture and Livestock Development
- Australian High Commission
- Australian Centre for International Agricultural Research
- International Livestock Research Institute
- Ballmer Group

We also extend our sincere appreciation to the AABCF Fellows from Kenya, Tanzania, Ethiopia, Cambodia, and Laos, whose participation enriched the discussions with practical perspectives from diverse livestock production systems.

Special thanks are due to the facilitators, technical experts, and organizers from the International Livestock Research Institute and partner institutions for their leadership in convening this important consultation and fostering collaboration across regions.

The commitment and collaborative spirit demonstrated by all participants during the workshop underscore the importance of partnerships in strengthening livestock research, innovation, and capacity development across Africa and Asia.

Cite this workshop report as follows:

Recha, J., Symekher, L., Whitbread, A. & Yao, N., (2026). Africa-Asia Biosciences Challenge Fund (AABCF) Eastern African workshop: Advancing climate-smart livestock practices through science and partnership. International Livestock Research Institute.

Photo 3: Group photo of delegates from ACIAR, ILRI, the University of Queensland, the University of Melbourne, and the AABCF team including fellows

ANNEX 1: Welcome and closing remarks

The workshop opened with a warm welcome from Dr. Nasser K. Yao (BecA-ILRI Hub), who set the overall context and purpose of the gathering. He welcomed participants to the Asia-Africa Biosciences Challenge Fund (AABCF) workshop, underscoring that livestock systems sit at the intersection of food security, livelihoods, climate change, and environmental sustainability. He noted that climate change is no longer a future concern but a present reality already affecting livestock producers across Africa and Asia through climate variability, emerging diseases, and declining productivity.

Dr. Yao highlighted that the AABCF builds on the strong foundation of the Africa Biosciences Challenge Fund (ABCF) while expanding its scope to include Asia, thereby strengthening South–South collaboration and broadening the reach and relevance of biosciences solutions. He outlined the structure of the five-day workshop, which progresses from setting the scientific and systems foundation, through advanced genetics, nutrition, and climate-smart production approaches, to field-based learning and fellow showcases. He emphasised the importance of partnerships, grounded and scalable solutions, and active engagement among fellows, supervisors, national institutions, universities, and the private sector.

Building on this foundation, **Prof. Appolinaire Djikeng (Director General, ILRI)** (2nd from the left) welcomed participants on behalf of ILRI and partners, acknowledging government representatives, the Australian High Commission, ACIAR, Australian university partners, CGIAR colleagues, and the BecA-ILRI Hub team. He recognised the resilience of the research and development community and reaffirmed the central role of livestock in the economies and food systems of countries such as Kenya, Ethiopia, and Tanzania. Prof. Djikeng stressed that the impacts of climate change are already evident at farmer level, reinforcing the urgency of the discussions taking place.

He linked the workshop to ILRI's 2030 Research, Innovation and Impact Strategy, which prioritises climate adaptation and mitigation through science-based, adoptable solutions. He highlighted ILRI's Livestock and Climate Solutions initiatives as platforms for packaging innovations that respond directly to country needs. Prof. Djikeng emphasised that the workshop is not only about advancing science, but also about prioritising research that responds to today's challenges, co-designing solutions with partners, and strengthening regional and intercontinental collaboration to ensure impact beyond borders.

Representing the Australian Government, **Mr. Christopher Ellinger (Deputy High Commissioner, Australia)** (second from the right) expressed appreciation to ILRI, CGIAR partners, government representatives, and all participants, particularly the fellows and experts who travelled long distances to attend. He reaffirmed Australia's strong commitment to people-centred development through long-term investment in capacity building via the Australian Centre for International Agricultural Research (ACIAR). He emphasised that while the programme focuses on climate-smart livestock systems and biosciences, its core purpose is to invest in people-scientists, practitioners, and future leaders, whose impact extends well beyond individual projects.

Mr. Ellinger highlighted the importance of strengthening national agricultural research and extension systems, noting that these institutions are critical in linking science to policy and innovation to farmers. He further emphasised the value of Africa-Asia-Australia collaboration, recognising that challenges such as climate change and food security are shared across regions. He commended the role of the BecA-ILRI Hub as a platform for African-led innovation, mentorship, and South-South learning, and expressed Australia's pride in supporting the AABCF initiative.

From a national policy perspective, **Dr. Christopher Wanga (Director, Livestock Policy and Research, Ministry of Agriculture and Livestock Development, Kenya)** (first from the right) conveyed the Government of Kenya's strong support for the workshop and its objectives. He emphasised that climate-resilient livestock systems are central to Kenya's food security, livelihoods, and climate adaptation agenda, particularly in arid and semi-arid lands where livestock is the backbone of local economies. He highlighted the severe impacts of climate change currently facing the sector, including drought, heat stress, water scarcity, disease pressure, and livestock losses.

Dr. Wanga outlined Kenya's progress in establishing climate-smart livestock policies, rangeland governance frameworks, and risk-informed planning systems, including early warning and drought preparedness mechanisms. He underscored a strategic shift from reactive, relief-based responses towards proactive, market-driven and asset-protection approaches that treat livestock as productive economic assets. He called for science- and data-driven solutions, strong public–private collaboration, and actionable recommendations that are implementable, financeable, and aligned with national policy and investment frameworks.

In closing, all speakers collectively encouraged participants to engage openly, share expertise, challenge assumptions, and build partnerships that extend beyond the workshop. The workshop was framed as a critical platform for co-creating scalable, context-specific solutions that strengthen livestock resilience, protect livelihoods, and contribute to food security and sustainable development across Africa and Asia.

Workshop Closing Remarks by Mr. Moses Kimani (Ministry of Agriculture and Livestock Development)

Mr. Kimani delivered the official closing speech on behalf of the Ministry.

- Support: He reiterated the Ministry's strong relationship with ILRI and expressed interest in integrating the fellows' research into national projects like the Kenya Livestock Commercialization Project (KELCOP).
- Advice to Fellows: He encouraged the fellows to leverage the network they had built during the week. His final advice on innovation was: "Fail fast, learn, adapt, and go back to the drawing board."

The workshop concluded with a group photograph and a commitment to maintain the network through digital platforms.

Photo 4: Distinguished Guests at the Opening Session

ANNEX 2: AABCF Workshop Agenda

Monday 19th January 2026

Day 1	**Theme: "Foundations of Climate-Smart Livestock: Productivity, Health and Data-Driven Improvement"**
Morning –	
09:00 – 10:00	**Welcome and introduction of the participants/objectives of the workshop** *Nasser Yao (BecA-ILRI Hub)* **Welcome remarks** *Appolinaire Djikeng (ILRI)-15 min* *Australian High Commissioner/Representative-15min* *Director, State Depart of Livestock, Kenya Ministry of Agriculture and Livestock Development-15min*
10:00 – 10:30	What is climate-smart livestock (CSL)? How does CSL link to other sustainable development themes? *Examples of climate change case studies in East Africa and Southeast Asia* *Anthony Whitbread and Claudia Arndt (ILRI)*
10.30–11:00	Group Photo and Tea/Coffee break *(30 min)*
11:00– 11:30	Overall livestock productivity framework and measures of performance *University of Queensland, University of Melbourne*
11:30– 13:00	Round-table discussion (*ILRI, UoM, UQ*)
13:00 – 14:00	*Lunch & break (1 hr)*
Afternoon –	
14:00 – 14:45	Climate smart agriculture in the livestock context *University of Melbourne*
14:45-15.30	Climate change/disasters and animal health *University of Melbourne*
15:30-16:00	Tea/Coffee break *(30 min)*
16:00 – 16:45	Data management basics and summarizing livestock performance (Excel, EBVs, replication) *University of Queensland*
16.:45 – 17.30	Animal health and productivity resilience problem solving *University of Melbourne*

Tuesday 20th January 2026

Day 2	**Theme: "Optimizing Livestock Nutrition, Genetics and Management for Climate-Smart and Low-Emission Production Systems"**
Morning –	
08:30 – 9:15	Climate change and environmental impacts on animal performance *University of Queensland, University of Melbourne*
9:30 – 10:00	Climate resilience small ruminant (sheep and goat) production *University of Melbourne*
10.00 – 10.45	Nutritional interventions for amelioration of heat stress *University of Melbourne*
10:45 – 11:15	Tea/Coffee break *(30 mins)*
11:15 – 12:00	Nutritional requirements & grazing management *University of Queensland*
12:00 – 12:45	Genetics & selection – tools & approaches (Genetics and Selection-Breeding climate resilient livestock) *University of Queensland*
12:45 – 13:30	*Lunch & break (45 min)*
Afternoon –	
13:30 – 15:00	Nutritional calculations (maintenance, growth, pasture moisture correction) *University of Queensland, University of Melbourne*
15:00 -15:30	Tea/Coffee break (30 mins)
15:30 – 17:00	Round-table discussion *(ILRI, UoM, UQ)*

Wednesday 21st January 2026

Day 3	Day trip to Kapiti Research Station and Wildlife Conservancy: Adaptation & resilience in livestock systems
Objective: This field visit will demonstrate innovative, science-based approaches to climate-smart livestock production, focusing on systems that enhance productivity and resilience while reducing greenhouse gas emissions in semi-arid rangelands. **Kapiti Research Station:** It is located approximately 60 km southeast of Nairobi and comprises 13,000 hectares of pristine semi-arid rangeland managed by the International Livestock Research Institute (ILRI). The station supports several thousand head of livestock, including cattle, goats, sheep, and camels, managed under extensive and climate-variable conditions. Kapiti serves as a living laboratory for research on climate-smart livestock systems, integrating animal health, productivity, genetics, rangeland and environmental sciences, forage production, and climate change mitigation and adaptation. The station also hosts a rich diversity of wildlife, providing a unique setting to study livestock–wildlife coexistence within sustainable rangeland ecosystems.	
07:00 – 07:30	Pickup from Hotel, and departure to Kapiti Research Station

09:00 – 09:30	Arrival at Kapiti. Refreshments, morning tea at homestead. Welcome and introductions to the ILRI Research team
09:30 – 11:00	**Technical presentations (Farmhouse):** • Climate-smart feeds and forages for resilient livestock systems – *Chris Jones* • Animal health and productivity in climate-variable environments – *Bernard Bett* • Genetic improvement for climate-smart livestock production – *Nehemiah Kimengich* • Measuring methane emissions from grazing livestock – *Nelson Kipchirchir*
11:00 – 13:00	**Tour and stops in the open field:** • Climate-smart feeds and forages for resilient livestock systems – *Chris Jones* • Animal health and productivity in climate-variable environments – *Bernard Bett* • Genetic improvement for climate-smart livestock production – *Nehemiah Kimengich* • Measuring methane emissions from grazing livestock – *Nelson Kipchirchir*
13:00– 13:30	Drive back to the homestead
13:30– 14:30	Lunch
14:30 – 15:30	Conservancy concept and management of wildlife and domestic livestock – *Ilona Gluecks*
15:30	Departure for Nairobi *(takes 1hr 30 mins)*

Thursday 22nd January 2026

Day 4	**Theme "Building Climate-Resilient and Sustainable Livestock Systems for Improved Animal Health and Productivity"**
Morning –	
08:30 – 10:30	Tools and techniques for resilience preparation: geographic Information Systems – the key things you need to know; spatial data (raster, vector) formats; and predictive modelling of vector-borne diseases. *University of Melbourne*
10:30 – 11:00	Tea/Coffee break *(30 mins)*
11:00 – 12:30	Methane production & intensity *University of Queensland, University of Melbourne*
12:30 – 13:30	Lunch & break *(1 hour)*
Afternoon –	
13:30 – 15:00	Round-table discussion *(ILRI, UoM, UQ)*
15:00- 15:30	Tea/Coffee break (30 mins)
15:30 – 17:00	Mazingira Centre visit: For awareness and learning on the process of greenhouse gas measurements and what this entails. *Claudia Arndt (ILRI)*

Friday 23rd January 2026

Day 5	**Showcasing Research and Building Technical Capacity for Resilient Livestock Solutions**
Morning –	
8:30 – 9:30	AABCF Research Fellows session: presentation of research topic, findings & application in current work institutions
9:30 – 10:30	AABCF Research Fellows session: presentation of research topic, findings & application in current work institutions
10:30 – 11:00	Tea/Coffee break *(30 min*)
11:00 – 12:00	AABCF Research Fellows session: presentation of research topic, findings & application in current work institutions
12:00 – 13:00	AABCF Research Fellows session: presentation of research topic, findings & application in current work institutions
13:00 – 14:00	Lunch & break *(1 hr)*
Afternoon	
14:00-15:30	Round-table discussion *(ILRI, UoM, UQ)*
15:30 – 16:00	Tea/Coffee break *(30 min*)
16:00-17:00	Plenary Discussions, Wrap-up and close (ILRI, UoM, UQ)

ANNEX 3: Participant List

	Name	Gender	Institution	Country
1	Dr. Michael Abera Hareri	M	Haramaya University	Ethiopia
2	Dr. Moses Mucugi Njire	M	Jomo Kenyatta University of Agriculture & Technology	Kenya
3	Dauson Felix Kamwangire	M	Tanzania Livestock Research Institute	Tanzania
4	Liberatus Katabazi	M	Tanzania Livestock Research Institute	Tanzania
5	Andrew Chota	M	Tanzania Livestock Research Institute	Tanzania
6	Erick Kimaro	M	Tanzania Livestock Research Institute	Tanzania
7	Vaileth Nyangwa	F	Tanzania Livestock Research Institute	Tanzania
8	Dr. Edward Moto	M	University of Dodoma	Tanzania
9	Dr. Anibariki Ngonyoka	M	University of Dodoma	Tanzania
10	Dr. Aziza Konyo	F	University of Dodoma	Tanzania
11	Dr. Zewdu Edea	M	Bio Emerging Technology Institute (BETin)	Ethiopia
12	Dr Eden Tesfaye	F	Haramaya University	Ethiopia
13	Caleb. S. Barasa	M	Kenya Agriculture and Livestock Organization	Kenya
14	Michael Baaria M'imunya	M	Kenya Agriculture and Livestock Organization	Kenya
15	Sally B. Nduta	F	Kenya Agriculture and Livestock Organization	Kenya
16	Prof (Eng.) James Messo Raude	M	Jomo Kenyatta University of Agriculture & Technology (JKUAT)	Kenya
17	Dr. Felix Kibegwa	M	University of Nairobi	Kenya
18	Dr. Oscar Koech	M	University of Nairobi	Kenya
19	Eng. Martin Mono	M	Kenya Meat Commission	Kenya
20	Dr. Stacey Waudo	F	Kenyatta University	Kenya
21	Surinder Singh Chauhan	M	University of Melbourne	Australia
22	Richard Eckard	M	University of Melbourne	Australia
23	Bereket Zeleke Tunkala	M	University of Melbourne	Australia
24	Abdul Jabbar	M	University of Melbourne	Australia
25	Kristy Digiacomo	F	University of Melbourne	Australia
26	Mark Stevenson	M	University of Melbourne	Australia
27	Frank Dunshea	M	University of Melbourne	Australia
28	Vitor Mercadante	M	University of Queensland	Australia
29	Kieren McCosker	F	University of Queensland	Australia
30	Christie Warburton	F	University of Queensland	Australia
31	Daniel Kapsandui Malinga	M	AABCF Fellow	Kenya
32	Willis Abwao Adero	M	AABCF Fellow	Kenya

	Name	Gender	Institution	Country
33	Beatrice Anyango Nabwire	F	AABCF Fellow	Kenya
34	Rosemary Peter Mramba	F	AABCF Fellow	Tanzania
35	Kabuni T. Kabuni	M	AABCF Fellow	Tanzania
36	Mussa Ngassa	M	AABCF Fellow	Tanzania
37	Elias Cherenet	M	AABCF Fellow	Ethiopia
38	Chernet Woju	M	AABCF Fellow	Ethiopia
39	Genet Zewdie	F	AABCF Fellow	Ethiopia
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